

WHAT IS AN MRI PROTOCOL?

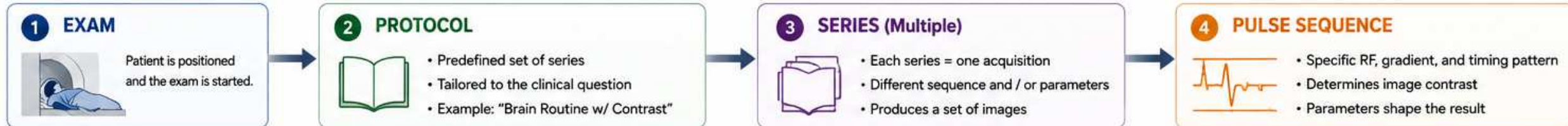
FROM EXAM TO IMAGES: PROTOCOL → SERIES → PULSE SEQUENCE

An MRI exam is not just one image. It is a collection of many series, each using a different pulse sequence to highlight different tissue properties. The technologist selects a protocol that answers the clinical question.



KEY CONCEPT

A protocol is a blueprint. Each series in the protocol uses a specific pulse sequence and set of parameters to create images with different contrast and clinical information.



EXAMPLE: BRAIN ROUTINE WITH CONTRAST



Clinical question:
Headache, dizziness, rule out lesion / stroke / demyelination

Protocol may include:

- Pre-contrast series
- Post-contrast series
- Localizer and calibration scans
- Multiple planes and thicknesses
- Optional advanced series (e.g., DWI, SWI, MRS)

TYPICAL SERIES IN A BRAIN ROUTINE PROTOCOL (EXAMPLE)

Series #	Series Name	Pulse Sequence	Plane	Purpose / What It Shows	Key Parameters (Examples)
1	Localizer	Localizer	3-Plane	Planning and prescription	-
2	T1W 3D	3D IR GRE (e.g., BRAVO)	Sagittal	Anatomy, detail, fat is bright	TR: ~8 ms TE: ~3 ms TI: ~450 ms FOV: 24 cm Slice: 1.0 mm
3	T2W FSE	FSE (T2)	Axial	Fluid and edema bright	TR: ~4000 ms TE: ~90-120 ms ETL: 16 Slice: 4 mm
4	FLAIR	IR-FSE (FLAIR)	Axial	Suppress CSF, lesions stand out	TR: ~9000 ms TE: ~90-120 ms TI: ~2500 ms Slice: 4 mm
5	DWI / ADC	EPI (DWI) + ADC Map	Axial	Detect acute ischemia / diffusion restriction	b-values: 0, 1000 TR: ~3000-6000 ms TE: ~70-100 ms Slice: 5 mm
6	SWI	3D GRE	Axial	Blood products, calcifications, venous structures	TR: ~28 ms TE: ~20 ms Flip: 15° Slice: 2 mm
7	Post Contrast T1W	3D IR GRE	Sagittal / Axial	Enhancing lesions, breakdown of blood brain barrier	TR: ~8 ms TE: ~3 ms TI: ~450 ms Slice: 1.0 mm

WHY SO MANY SERIES?



T1-Weighted

Best for anatomy, fat, subacute bleed, post-contrast enhancement.



T2-Weighted

Best for fluid, edema, and many pathologies.



FLAIR

Suppresses CSF to detect lesions near ventricles.



DWI / ADC

Detects early stroke and cellular changes.



SWI

Sensitive to blood, calcification, and venous structures.



Post Contrast

Highlights areas with abnormal blood brain barrier or enhancement.



Each series answers a different clinical question. Together they provide a complete picture.

ENGINEER VIEW WHAT TO KNOW



KNOW THE PROTOCOL

- Understand what protocol was run.
- Know typical series order.
- Ask for the protocol name and version.



COMPARE, DON'T ASSUME

- Compare with a known good exam.
- Same protocol + different result = investigate.



CHECK VARIABLES

- Different coil?
- Different patient?
- Different parameters?
- Different plane?



LOOK FOR PATTERNS

- One series only?
- All series?
- One plane or all planes?
- Pre or post contrast only?



COMMUNICATE

- Talk with the technologist.
- Understand what they adjusted.
- Ask what changed.



THINK SYSTEMATICALLY

- Patient → Coil → Protocol / Parameters → Prescan → Image → Hardware
- Don't jump to hardware.



The technologist chooses the right sequence for the right reason. The engineer understands how and why it works — and when it doesn't.